

Lab 3D: Capstone - Multi-Agent Build + Headless CI Check

Duration: 35 min

After: Scale, Cost & New Surfaces lecture (Day 3, Block 4)

Prerequisites: Labs 3A-3C complete (MCP server, subagents, ci-review script)

Objective

Run everything at once: a multi-agent build of a real feature on your toolkit repo, graded against a rubric, validated by your headless CI check. Success is binary: **toolkit repo complete + CI run green**.

Deliverable (toolkit chain 12 of 12 - FINAL)

The finished personal toolkit repo containing all five artifacts - **CLAUDE.md**, **skill**, **hook**, **command**, **plugin manifest** - plus a new feature built by agents and a green CI run proving it.

START MARKER: `cd business-dashboard`, *terminal open*.

Fell behind? Minimum viable start:

Set up this project with: CLAUDE.md conventions, an Express server with `src/routes/`, a code-review skill in `.claude/skills/`, a PostToolUse prettier hook in `.claude/settings.json`, a `/review` command, subagents `security-auditor` and `grader` in `.claude/agents/`, and `scripts/ci-review.sh` that runs `claude -p` with `--allowedTools "Read,Grep,Glob"` `--max-turns 10` and fails below score 6.

Phase 1: Toolkit Inventory Gate (5 min)

Before building, verify all five toolkit artifacts exist. Start `claude` and run:

```
Verify this repo contains all five toolkit artifacts and report PRESENT/MISSING for each:
1. CLAUDE.md (root, under 200 lines)
2. A skill: .claude/skills/code-review/SKILL.md with
   name/description/disable-model-invocation/allowed-tools frontmatter
3. A hook: .claude/hooks/post-edit-format.sh registered as PostToolUse in
   .claude/settings.json
4. A human-invoked skill: .claude/skills/review/SKILL.md using $ARGUMENTS with
   `disable-model-invocation: true`
5. A plugin manifest: ../dashboard-toolkit-plugin/.claude-plugin/plugin.json (or plugin
   installed from team-marketplace)
```

Expected output marker: five PRESENT lines. Fix any MISSING with the relevant lab's recovery prompt before continuing - the capstone grade depends on it.

Phase 2: Plan the Feature (5 min)

Shift+Tab to **Plan mode**, then:

Plan a "Weekly Report" feature:

- GET /api/report/weekly - JSON summary: metrics by category, task completion rate, top 3 high-priority open tasks
- A "Weekly Report" section on the dashboard UI rendering that data
- Data shapes should match the dashboard-metrics MCP server where relevant
- Must follow CLAUDE.md conventions and pass our rubric.md criteria

Produce a step-by-step plan with file list and test strategy. Save it to IMPLEMENTATION_PLAN.md when I approve.

Review, iterate once if needed, approve, and let it save the plan.

Expected output marker: IMPLEMENTATION_PLAN.md exists; no source files changed yet (Plan mode held the line).

Phase 3: Multi-Agent Build (12 min)

Shift+Tab back to Default (or Auto if your Lab 1D permission rules are in place), then orchestrate:

Execute IMPLEMENTATION_PLAN.md as a multi-agent build:

1. Spawn a builder subagent to implement the API endpoint and UI section per the plan
 2. In parallel, spawn a test-writer subagent to write tests for the endpoint (happy path, empty data, error handling)
 3. When both finish, run the security-auditor subagent over the changed files
 4. Finally, have the grader subagent score the implementation against rubric.md
- Report each agent's summary and the grader's verdict.

Watch: builder and test-writer run as background subagents in parallel (streaming since week 27); auditor and grader run as review passes.

Expected output marker: all four agent summaries + a grader table ending TOTAL: n/10. If the verdict is REWORK:

Fix only the criteria scored below 2, then have the grader re-score.

Expected output marker: PASS ($\geq 8/10$).

Verify by hand - open <http://localhost:3100> and confirm the Weekly Report section renders. Then:

```
/review src/routes/
```

Commit:

Commit the weekly report feature with a conventional message, mentioning it was built by parallel subagents and graded at [score].

Phase 4: Headless CI Check - the Green Light (8 min)

Exit Claude (exit). Run your Lab 3C gate against the finished repo:

```
./scripts/ci-review.sh
```

Expected output marker:

```
Review score: 8
Cost: $0.0x
REVIEW PASSED
```

Exit code 0 = **CI green**. If it fails, read the findings:

```
jq -r '.result' review.json
```

Start **claude**, paste the top issues with "Fix these findings, then re-run scripts/ci-review.sh", and repeat until green.

Built the CI pipeline in Lab 3C? Push a PR instead and let the pipeline post the advisory review - same gate, running on a server rather than your laptop:

```
git checkout -b feature/weekly-report && git push -u origin feature/weekly-report
```

Then open the PR:

```
# Azure DevOps
az repos pr create --source-branch feature/weekly-report --target-branch main \
--title "feat: weekly report (capstone)" --description "Multi-agent build, grader-verified, CI-checked"

# GitHub
gh pr create --title "feat: weekly report (capstone)" --body "Multi-agent build, grader-verified, CI-checked"
```

Watch the run in the Azure DevOps **Pipelines** tab, or with `gh run watch` on GitHub.

Phase 5: Retrospective (5 min)

Using the git history and current codebase, summarize the evolution of this repo across all 12 labs: what each toolkit artifact adds, how the agents divided today's work, and the total cost of this session (/cost data). Keep it under 300 words - this is my week-one plan seed.

Count your prompts across three days. A few dozen messages shipped: a working system, a policy file, two tested skills, enforced hooks, a distributable plugin, an MCP server, a graded multi-agent feature, and a CI pipeline. That's the leverage.

FINISH MARKER - Capstone Success Criteria (binary)

Complete = BOTH true:

- [] **Toolkit repo complete:** CLAUDE.md + skill + hook + command + plugin manifest all PRESENT (Phase 1 gate)
- [] **CI run green:** `scripts/ci-review.sh` exits 0 (or the GitHub Action run completes with the advisory review posted)

Supporting evidence:

- [] IMPLEMENTATION_PLAN.md produced in Plan mode
- [] Feature built by parallel subagents, not the main session
- [] Grader verdict PASS ($\geq 8/10$) with evidence table
- [] Weekly Report renders in the browser
- [] Conventional commit history tells the story

If Stuck

Problem	Recovery
Phase 1 shows MISSING artifacts	Run that lab's recovery prompt - do not skip the gate
Agents collide on the same file	Re-run with explicit ownership ("builder owns src/, test-writer owns tests/ only")
Grader keeps failing one criterion	Read its evidence column - fix that exact line, don't rewrite the feature
ci-review.sh score seems wrong	<code>jq -r '.result' review.json</code> and read the findings - the score follows them
Ran out of time	Binary criteria first: inventory gate + one green CI run beats a fancy feature
Context bloated	<code>/compact</code> , or rewind to the post-plan checkpoint and re-run Phase 3 tighter

Debrief Questions

1. Which toolkit artifact did the most work during the capstone - policy, skill, hook, command, or plugin?
2. Where did the grader catch something the builder believed was done?
3. What's your week-one plan: which artifact goes into your real repo Monday, and what's your CI entry point?